

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
CHANDPUR SCIENCE AND TECHNOLOGY UNIVERSITY

Course Code: CSE 2201

Course Title: Algorithm Design and Analysis

Time: 1 Hour

Full Marks: 20

Student ID:

Date:

CT-02

Answer any **Two** (2) from the following **Three** (3) Questions.

Q. No.		Marks
1.	A vending machine in a train station gives out change using coins of 1, 2, 5, and 10 Taka. For a return of 38 Taka, explain how the greedy algorithm can be used. What will be the output if the coin denominations are {1, 3, 4}? Does the greedy algorithm still work optimally?	10
2.	Consider the following graph with weighted edges: Vertices: A, B, C, D, E Edges: A-B (2), A-C (3), B-C (1), B-D (4), C-D (5), C-E (6), D-E (7) a) Find the Minimum Spanning Tree using Prim's Algorithm . b) Find the Minimum Spanning Tree using Kruskal's Algorithm .	10
3.	Consider the following characters and their frequencies: A: 5, B: 9, C: 12, D: 13, E: 16, F: 45 a) Construct the Huffman tree. b) Find the Huffman codes for each character. c) Calculate the total number of bits used to encode the string "ABACD".	10